INTERNET

The Internet is a global system of interconnected computer networks that use the standard Internet protocol suite (TCP/IP) to serve several billion users worldwide.

Source: https://en.wikipedia.org/wiki/Internet

WORLD WIDE WEB (WWW)

- The World Wide Web (WWW) was developed in the late 1990 by Tim Berners-Lee.
- WWW is a client-server model and uses TCP connections to transfer information or web pages from server to client.
- WWW uses a Hypertext model.
- Hypertext allows interactive accesses to a collection of documents.
- Documents can hold
  - Text (hypertext), Graphics, Sound, Animations, Video

INTERNET & WWW

- The Internet and the World Wide Web are *not* the same.
- The Internet is a global system of interconnected computer networks.
- In contrast, the www is one of the services that runs on the Internet.
- www is a collection of text documents and other resources, linked by hyperlinks and URLs, usually accessed by web browsers from web servers.
- In short, the web can be thought of as an application running on the Internet.

INTERNET & WWW

- The Internet works by defining an address of each resource attached to it. This is Internet Protocol (IP) address.
- Ex: 158.162.131.236
- We don't use numbers (IP)..!! We use names (e.g., www.yahoo.com).
- The name actually will have associated number (IP address).
- When name is entered into a browser, the computer will use a special system called Domain Name System (DNS) to look up the associated number.

HTTP PROTOCOL

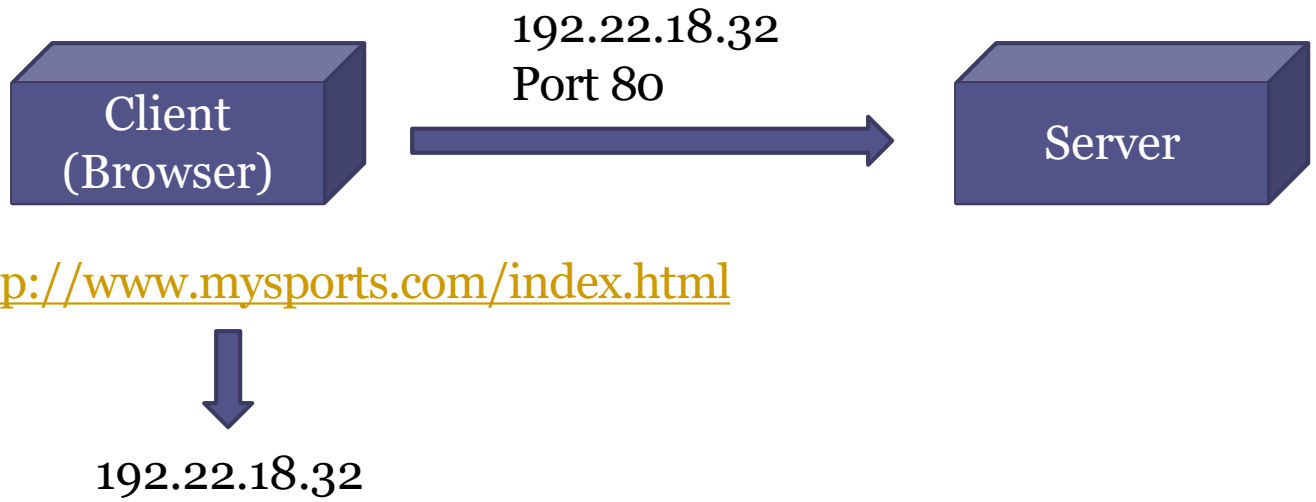
- The main protocol used for communication between a browser and a web server is HTTP.
- For web documents the HTTP protocol works by sending commands over a TCP connection.
- Main model used for the web is *client-server*.



http://www.mysports.com/index.html

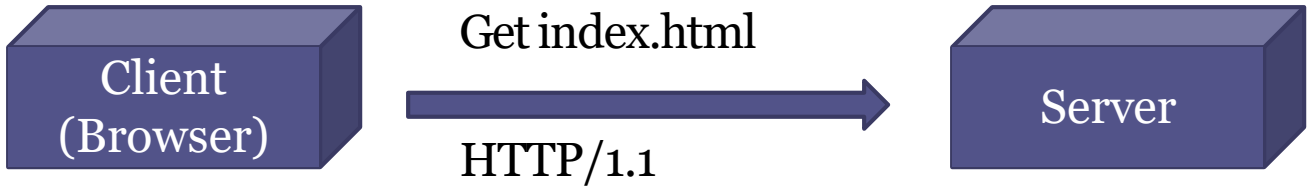
- First stage: user types a URL in the browser address window.

PROTOCOLS AND PROGRAMS: WEB PAGES



- Second stage: URL is converted into IP address, which is then used to make a connection to the server via port 80, the one used for HTTP and web.

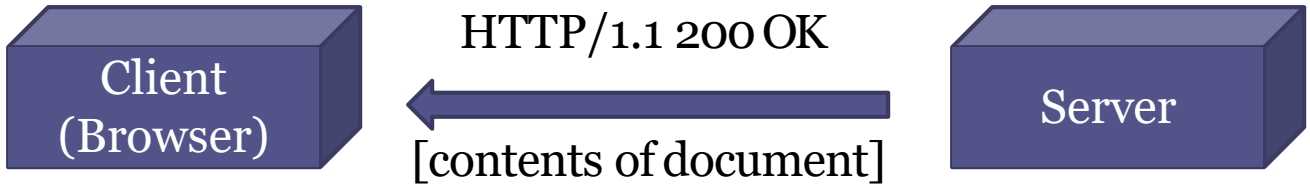
PROTOCOLS AND PROGRAMS: WEB PAGES



http://www.mysports.com/index.html

- Third stage: client extracts the file name from the URL and sends the ***request***. When received, the server looks up the request.

PROTOCOLS AND PROGRAMS: WEB PAGES



http://www.mysports.com/index.html

- Final stage: if all is well, the server sends an HTTP ***response*** message saying that the page was found, followed by the page itself. When the page has been sent the connection is dropped.

APPLICATIONS AND DEVELOPMENT TOOLS

- To start developing an application, simple text editor on your computer works.
- Browser
- FTP Client in case of uploading files to the server
- Editors specific to the languages
- Editors for HTML, JavaScript, PHP, Perl and most other web languages

WEB BROWSER

- One of the main tools used while developing web applications is the Web Browser.
- Browser connects over the internet to a web server.
- Uses HTTP protocol
- The format of incoming web pages is HTML which is interpreted by the browser and displayed.
- In addition to HTML, most browsers support other types of files
  such as JPEG, GIF and PNG.
- Many modern browsers work within the standard versions of HTML and XHTML.

FEATURES OF WEB BROWSER

- Main function of a browser is to show text and graphics and possibly other media.
- 'Preferences' or 'Tools' section of browser allows to customize or set up special options. (ex: set home page)
- 'Cache', a local memory area, speeds up loading of regularly visited pages, by storing a copy in the client computer, which it will show when the page is requested again.
- 'Clear the Cache' option allows to empty the cache memory at times.
- For example, while developing web pages, if the page is cached, it may not update when you refresh the current page view.
- We can turn caching off or put a cache size of 0.

FEATURES OF WEB BROWSER

- Preferences has option of cookies which are small pieces of information to identify you when you revisit the site.
- In some cases, you can switch off the acceptance of cookies.
- Both cached pages and cookies can be cleared from memory, either individually or all together.
- Auto filling of forms is other option if you want your details to be automatically added when the site gives you a form to fill out.
- Download location option allows you to select where the files you want to download will go.

CONCEPT OF EFFECTIVE WEB DESIGN

Guideline for Good Design:

- The website can be viewed on all the monitors.
- Fonts are of moderate size. The most commonly used fonts are Times New Roman, Arial and Verdana.
- The contrasting color are preferred. The color contrast should not be more than three colors.
- The font color should be consistent throughout the website. Typically at most two font colors are used.

CONCEPT OF EFFECTIVE WEB DESIGN

- The navigation from on page to another must be smooth.
- The web page should be loaded up in few seconds.
- Must be compatible in all the most commonly used browsers which are internet explorer, Mozilla Firefox, Google Chrome

CONCEPT OF EFFECTIVE WEB DESIGN

- **Guideline for Bad Design:**

- The size of web pages is not viewable in standard browser and the user has to set the pixel width and height manually for viewing the website.
- Use of big fonts, unattractive colors, big buttons and long text on the web pages.
- Horizontal scroll annoys the user.
- Due to heavy use of graphics, animation and pictures there is load time problem for the web page.
- Displaying the text with wrong spelling.

WEB SITE DESIGN ISSUES

- Browser
- Bandwidth and Cache
- Display resolution
- Look and Feel of the Website
- Page Layout and linking
- User centric design
- Sitemap

WEB SITE DESIGN ISSUES

- **Browser and Operating Systems**

- Challenged faced by the author of web document is designing the web pages which can be display properly by multiple browser.

- Different versions of HTML also support different sets of tags.

- Moving from HTML to XHTML.

- Older version of a browsers may not support recent tags and features.

WEB SITE DESIGN ISSUES

- Still follow the older versions of browsers although newer versions are available.
- Ex, if your page is designed with Cascading Style Sheet(CSS) and if the user browser does not support CSS, the page looks very different.
- So, the browser compatibility is a critical issue to make pages portable on the different browsers with different versions.
- Responsibility of a web designer is to test the web page of as many browsers as possible to ensure that the work appears correctly.

WEB SITE DESIGN ISSUES

- **Guideline for browser compatibility issues:**

- **Follow W3C standards:** Use of scripting language such as HTML, CSS, JavaScript correctly.
- **Validate the code:** Test the syntax of the code. Make the web document free from any kind of coding errors.
- **Know your Audience:** The web design must be accessible and readable for the target audience.
- **Test your work on multiple browsers and devices:** Test and try to execute the web document on multiple browsers and devices for ensuring its correctness.

WEB SITE DESIGN ISSUES

❑ Bandwidth and Cache

- Connection speed plays an important role in designing web pages.
- Bandwidth is used to define user's internet connection speed.
- If user has low bandwidth connection and web page contains too many images, it takes more time to download.
- Generally users have no patience to wait for longer time than 10-15 seconds and move to other site without looking at contents of your web pages.

WEB SITE DESIGN ISSUES

- Hence , while designing web pages, have to consider low speed users like dial-up connections (having 56-256 Kbps).
- High-speed connection with broadband technology has solved this problem significantly.
- With high- speed connections more attractive graphics can be used to design visually attractive pages.
- Browsers provide temporary memory called cache to store the graphics.

WEB SITE DESIGN ISSUES

- When user gives the URL of the web page for the first time, HTML file together with all graphics file referred in a page is downloaded and displayed.
- At the same time, the downloaded images are stored in cache area of browser for specified time as user's preference.
- When next time the same page is visited, the browser gets only HTML file and uses the images from cache which increases the downloaded speed significantly.
- This feature is very helpful specifically on low-speed connections

WEB SITE DESIGN ISSUES

- **Display Resolution**

- Measured in terms of pixels and common resolutions are 800 * 600 and 1024 * 786.
- With the reduction in cost of large size monitors , 1240 * 1024 resolutions will become common in near future.
- Most monitors are using resolution of 1024*768 pixels.
- Many old monitors are working on 800*600.

WEB SITE DESIGN ISSUES

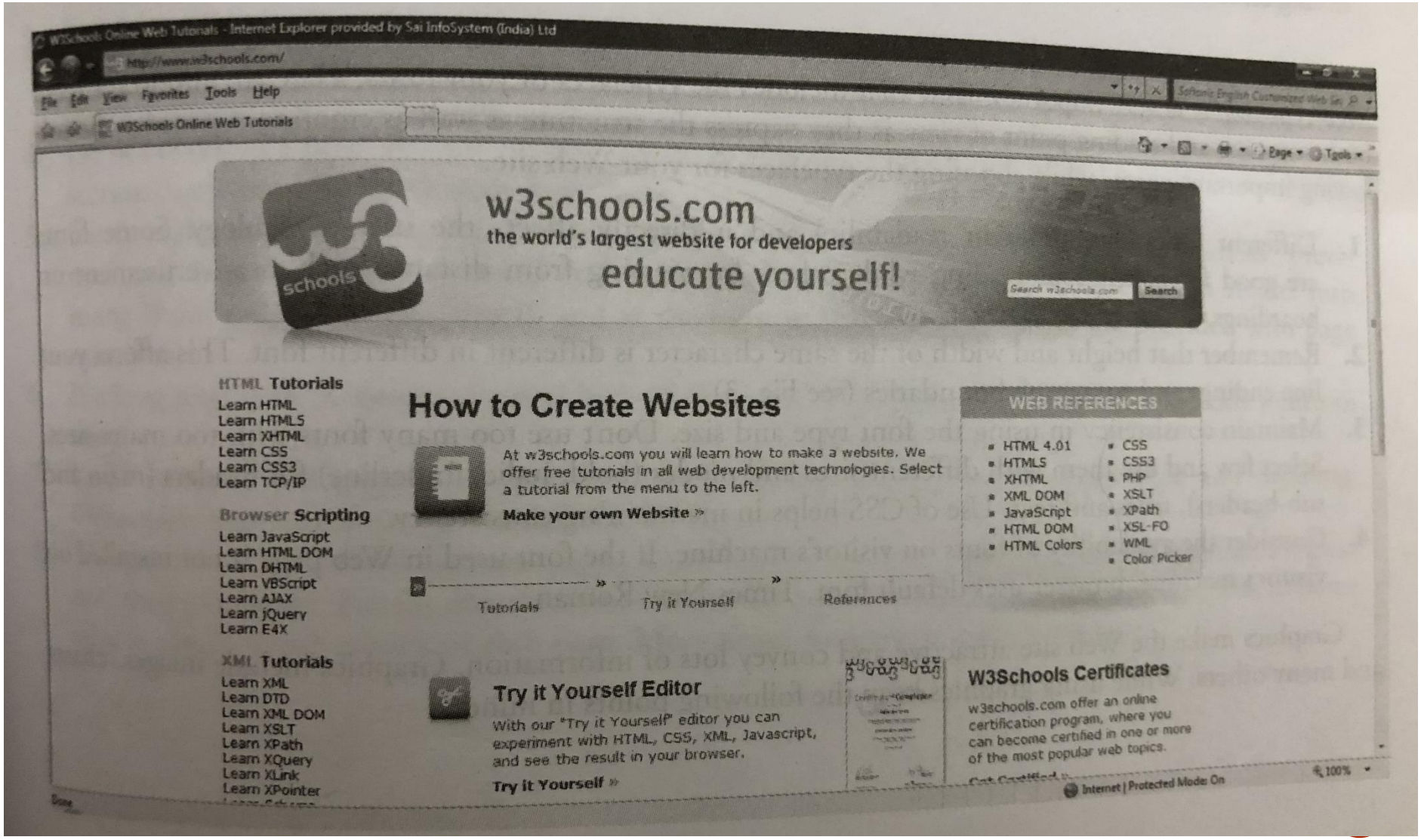
- Three choices for web page design.

- Design a web page with fixed resolution say 800*600.
- The web page properly fits into a screen with this resolution but leaves some part of the screen if resolution is 1024*786.
- If design is fixed to use 1024*768, it will not fit into the screen having 800*600 resolution.

WEB SITE DESIGN ISSUES

- Make a flexible design using HTML table to fit into different resolution.
- Web page is divided into three columns with middle column having a variable width.
- Depending on the resolution of monitor on which page it is viewed, it changes its width.
- If the page is displayed on a monitor with a higher resolution, the page displayed on left-hand side and some part on the right- hand side remains blank.
- We can use centered design to display page, e.g. in middle of the screen, leaving equal space on both side of the page.

WEB SITE DESIGN ISSUES



WEB SITE DESIGN ISSUES

❑ Look and Feel

- Look and fell of the web site decides the overall appearance of the web site.
- It includes all design aspects such as
  - Web site theme
  - Web typography
  - Graphics
  - Visual structure
  - Navigation, etc.

WEB SITE DESIGN ISSUES

❖ Web Site Theme

- All pages of the Web site hold together and give the impression of a single unit.
- Web site themes gives the impression to the users that they are on same Web site while navigating through various pages.
- Every organization has unique reason to create their Web site.
- Web site theme should reflect the objective of the organization and convey the message of the organization.

WEB SITE DESIGN ISSUES

- Some examples of the way we can choose the theme for a Web site:

- Use logo of the company as theme. It must appear on each page to convey your message.
  - For example, Telecommunication Company can use well-design logo and pictures though out the site.

- Use color scheme as theme. In such case, all your buttons, links, titles, labels and other text should use it consistently.

- Web site for global warming can use pictures, message related to cause and effect of the global warming on lives.



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WEB SITE DESIGN ISSUES

- **Fonts, Graphics and Colors**

- Web typography is very important issue that includes the typefaces of your fonts.
- They are important from both scanning and reading point of view as they express the structure as well as emotion.

WEB SITE DESIGN ISSUES

- **Following important points while choosing the typefaces for your Web site.**

- Different fonts have different readability and it directly affects the user psychology. Some fonts are good for continuous reading while others for viewing from distance such as advertisement on hoardings.

- The height and width of the same character is different in different font. This affects your line endings and paragraph boundaries.

- Maintain consistency in using the font and size. Don't use too many fonts with too many sizes.

WEB SITE DESIGN ISSUES

- Select few and use them with different sizes and modes (bold, italic, underline) for headers (main and sub-headers), titles and text. Use CSS helps in maintaining consistency.

- Consider the availability of fonts on visitor's machine. If the font used in Web page is not installed on visitor's machine, browse uses default font, Times New Roman.

- Graphics make the Web site attractive and convey lots of information.

- Graphics include images, charts and many others.

WEB SITE DESIGN ISSUES

- **Following important points while using graphics**

- One of the most important points is the graphics file format. Different file formats support different levels of the compression and other features.
- Popular image formats on Web are JPG, GIF, PNG, etc. Use them at right place considering their features.
- Too many graphics or images with larger sizes reduce the download speed. Use limited graphics or images with smaller size or in compressed format.
- Use images that suit the theme of the Web site and profile of target audience rather using any image just to include graphics.

WEB SITE DESIGN ISSUES

- Colors are part of life and Web sites are not exceptions.
- Web site with good content but improper color scheme does not attract viewers and hence it is fails.
- Color scheme that suits your theme, content and liking of your target audiences plays a critical role.
- For the graphic and color schemes, help of graphic and art designers is required to give the Web site commercial and professional touch.

WEB SITE DESIGN ISSUES

❖ Presentations and Access

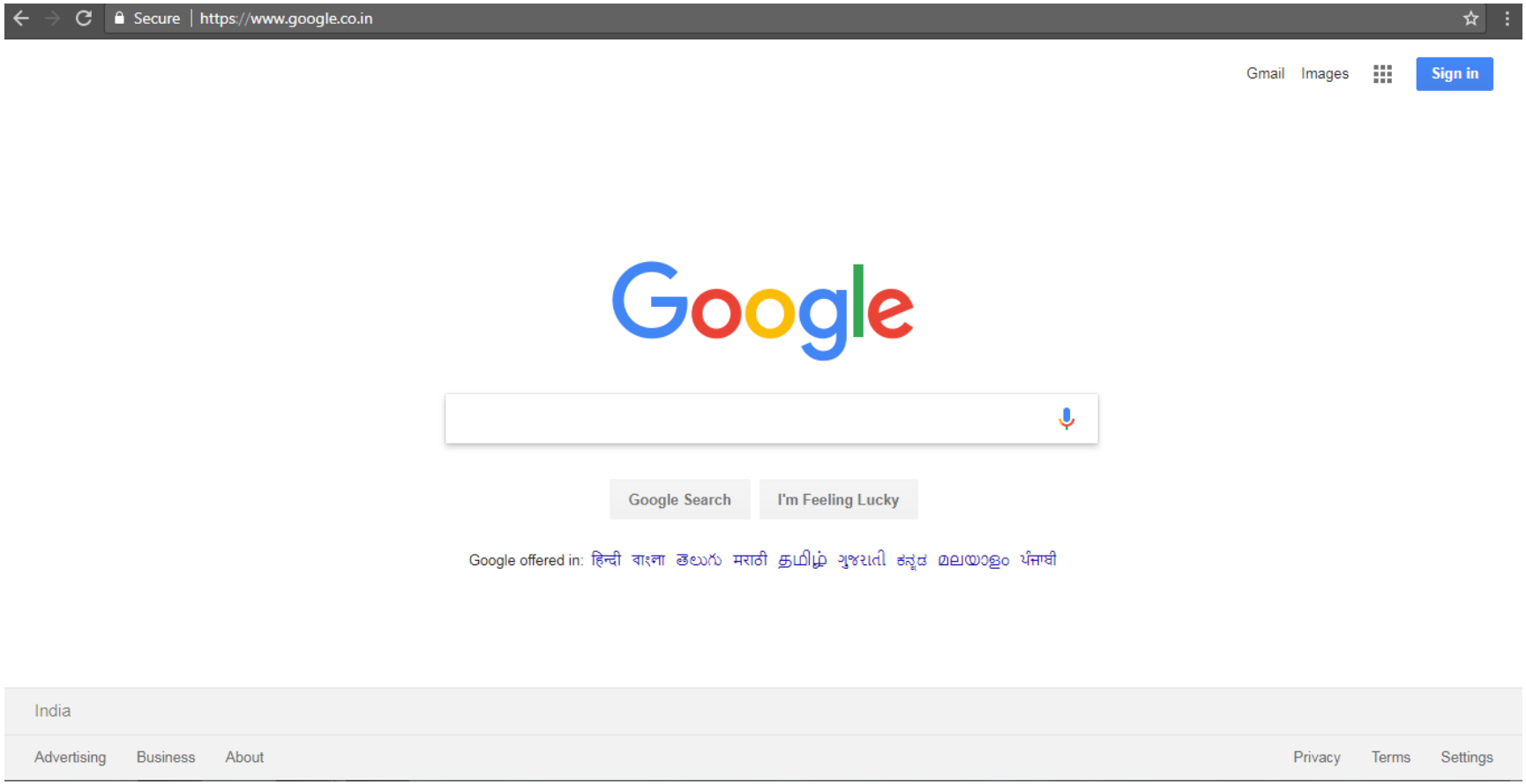
- Websites are designed to provide contents online to the diverse group of visitors having different reading habits and expectations.
- A clear presentation helps in accessing the desired information quickly and easily without any hurdles.
- The visual structure of Web page, navigation links and use of the white spaces decides your presentation clarity.

WEB SITE DESIGN ISSUES

- **Consider the following points**

- Web pages should be divided in visually different areas to present the contents with different importance. Use grid-based structure to divide the Web page into different rows and columns.
- Make different parts of the page are visible by using the white space, i.e. blank areas between them to make the contents eye- catching.
- Keep the page simple and focused if required.
- Do not overload a page with too much content. Divide the contents into parts, i.e. sections and within section pages and link them using hyperlinks.

WEB SITE DESIGN ISSUES



WEB SITE DESIGN ISSUES

- Don't make page too lengthy. Instead, divide it into multiple pages with links such as "previous" and "Next" to move forward and backward.
- If a long page is to be designed, provide the link to top of the page at bottom to quickly return to start.
- Links between the pages and sections should exhibit consistency in appearance and meaning. Otherwise, that makes navigation difficult for the visitors.
- Navigation, i.e. transitions between pages, should reflect the consistency in terms of graphics for links and their placement.
- Put the common links at the same place on each page.

WEB SITE DESIGN ISSUES

❑ Page layout and Linking

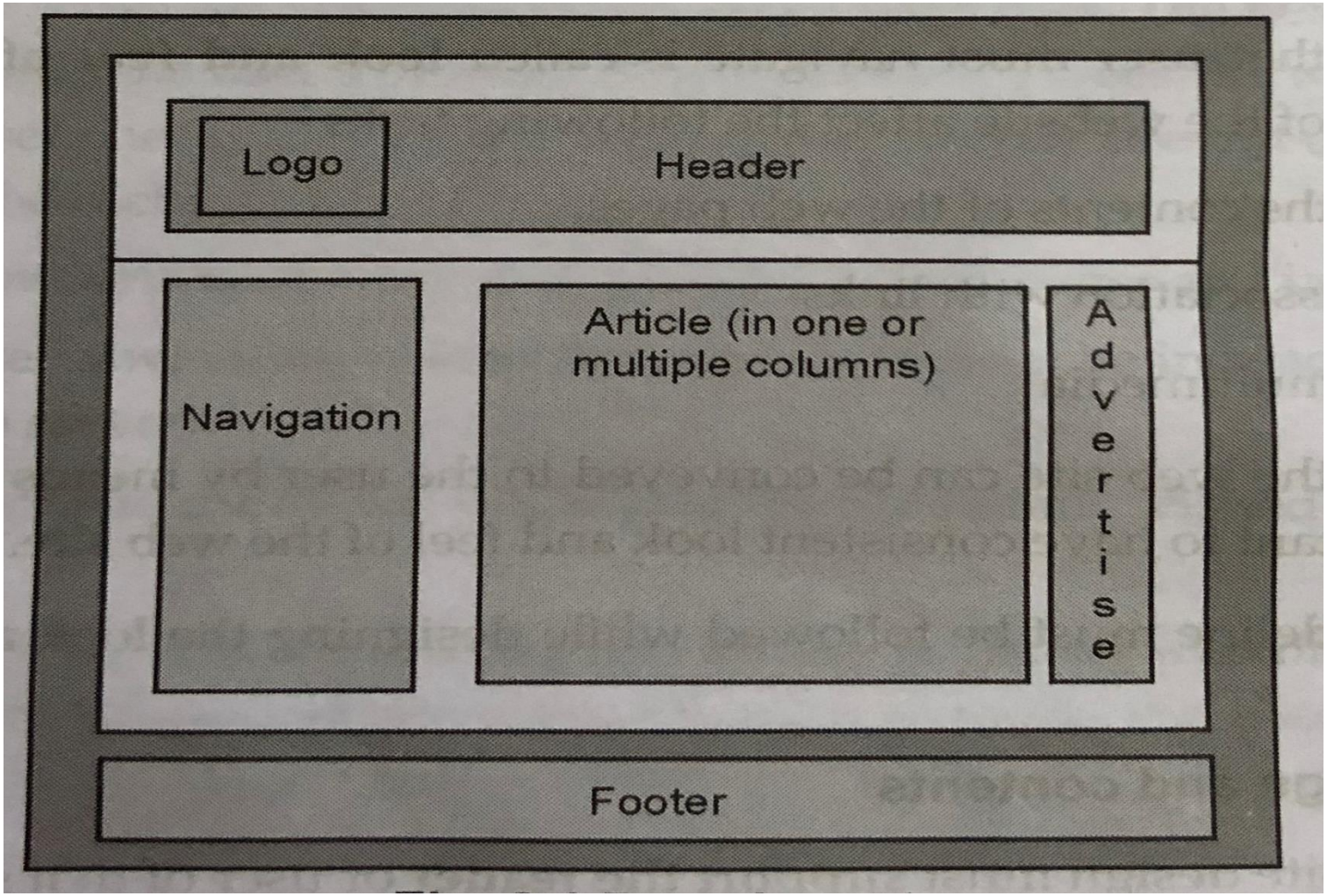
- Web sites consists of individual Web pages that are linked together using various navigational links.
- Page layout defines the arrangement of content on the page with some specification or standard.
- Allows the designer to distribute the contents in a page such that visitor can view it easily and find necessary details.
- Page layout is also called the page template.
- Readymade templates are available on the Web sites which can directly used to design the pages quickly by adding the contents into it.

WEB SITE DESIGN ISSUES

- **Consider the following points while designing layout**

- While distributing the contents in layout, balance between text and graphics. Too many graphics may lead to slow download of the page.
- Locate the items on the page using screen importance to attract the user's attention on important items.
- Maintain consistency among the layout of the pages.
- Once the page layouts are prepared, we have to link them together to provide the navigation paths consistently so that user can find desired information quickly and easily.

WEB SITE DESIGN ISSUES



WEB SITE DESIGN ISSUES

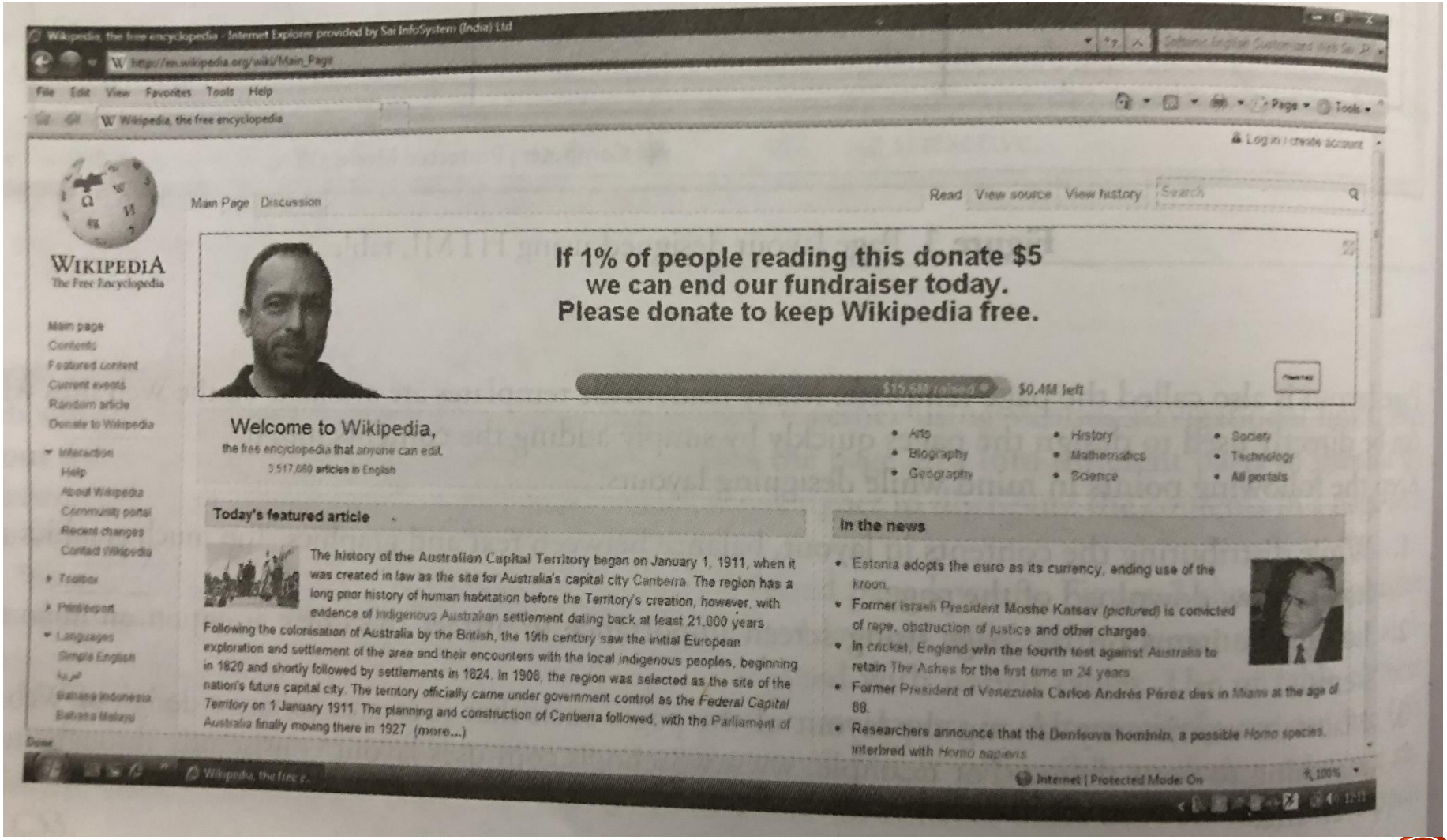
Locating Information

- Web page is viewed on a computer screen and the screen can be divided into five major areas such as center, top, right, bottom and left.
- Put most important information in the center to get the attention of the user.
- Well-designed Web sites use left- hand side for providing links to different items or sessions in form of menu.
- Top is used for displaying logo and title as well as flashing news links or at many times advertisement.

WEB SITE DESIGN ISSUES

- Right-hand side is used for links to other important information.
- Bottom is used for quick links to important sections or to provide copyright message.

WEB SITE DESIGN ISSUES



WEB SITE DESIGN ISSUES

❑ Make Design User-Centric

- Difficult for any Web designer to predict the exact behavior of the web site users.
- User- Centric design means all the design efforts are made by keeping user as a main important factor.
- Idea of general behavior of common user helps in making design of the Web site user-centric.
- What are the expectations of the user from is site is found out
- If the information is provided for reading, e.g, article on an online magazine or newspaper site, then user's general reading habit gives the idea of how to organize such information.

WEB SITE DESIGN ISSUES

- People normally read from left to right and top to bottom. Hence, we can organize such contents in column fashion on a Web page.
- Article or tutorials divided into sections, the table of contents greatly helps user to quick access the desired part of tutorial.
- The links in table of contents change colors once visited, so user gets an idea of which links have been visited and which are yet to be visited.
- Web pages are linked together to provide non-sequential but quick reading.
- It sometimes necessary to highlight the words in a paragraph as link to provide the links to the pages giving detail meaning of the word.

WEB SITE DESIGN ISSUES

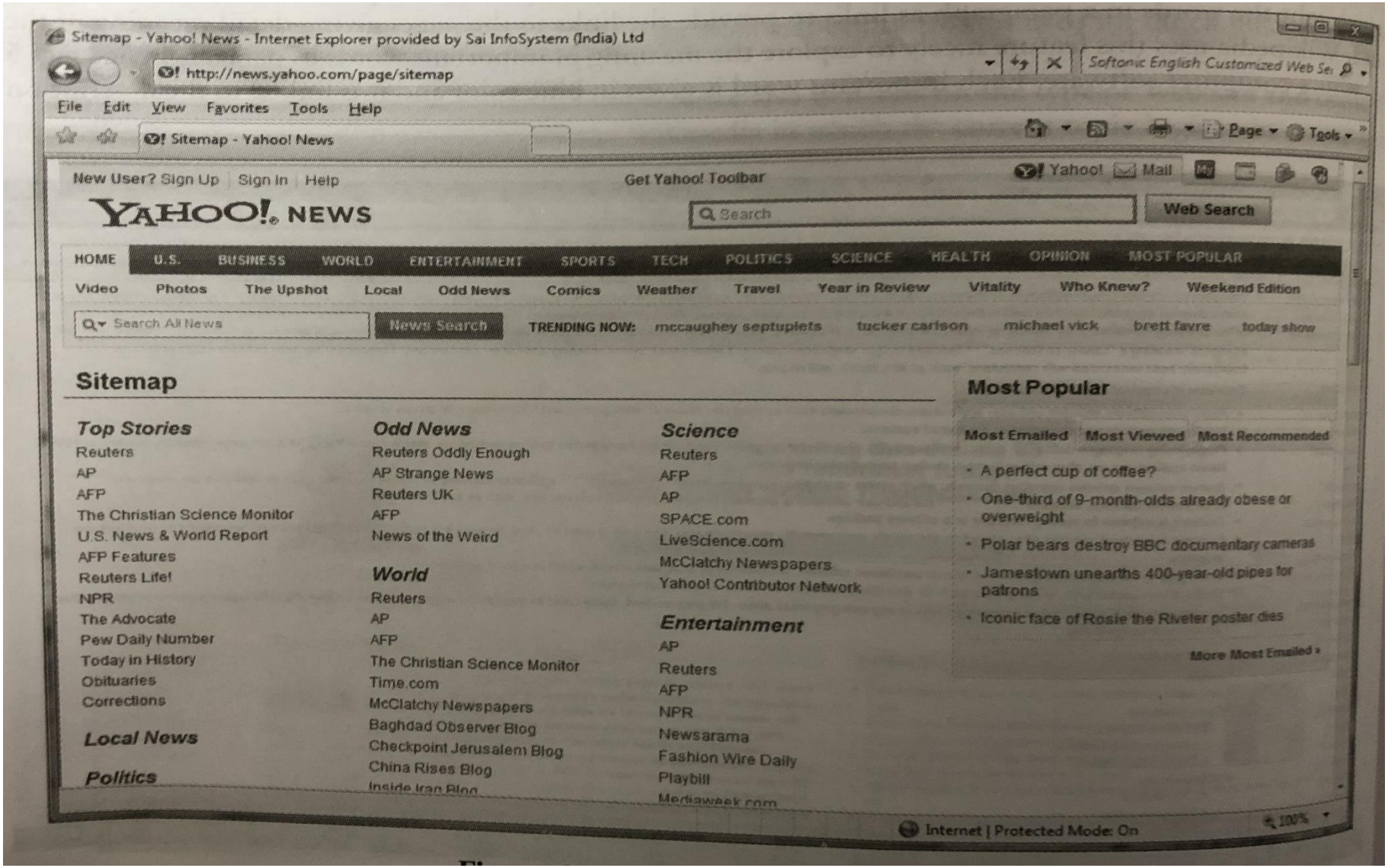
❑ Sitemap

- Many times Web sites are too complex as there are a large number of sections and each section contains many pages.
- It becomes very difficult for a visitor to quickly move from one part to another.
- Once the user selects a particular section and pages in that section, the user gets confused about where he/she is and where to go from there.
- To make it simple, keep your hierarchy of information to few levels or provide the navigation bar on each page to jump directly to a particular section.

WEB SITE DESIGN ISSUES

- Another solution is to provide the sitemap including links to each section and their pages directly.
- Sitemap gives the organization of content of a web site graphically.
- By, clicking any link on sitemap, you can directly jump to that page.

WEB SITE DESIGN ISSUES



PLANNING A WEB SITE

- To achieve higher success of the Web site in terms of user satisfaction, better planning is needed.
- Before start developing a Web site, should ask questions such as
  - ✓ Why are we developing this Web site?
  - ✓ What do we achieve by developing this website?
  - ✓ Who are the people who will use this website?
  - ✓ What are the information contents?
  - ✓ How are these contents organized? What are the possible way?
  - ✓ How the files prepared are organized.

PLANNING A WEB SITE

- The project planning phase is carried out in following steps:
  - ✓ Deciding the goals and objectives, clients expectations and user's need.
  - ✓ Focusing the group for which the web project has to be developed
  - ✓ Perform the market research
  - ✓ The concept development can be done by creating the prototypes.
  - ✓ Then the creative design is created so that the developer can communicate with the user for understanding their requirement.

PLANNING A WEB SITE

❑ Objective and Goal

- Many times activity fails or desired results are not achieved because of unclear objective or purpose
- If the objective for developing the Web site is clear, rest of steps in planning will become easier, otherwise it may create many issues.
- The Web Sites are developed for different reasons.
- It may be provide information and better service to customer, to update the members in a community.
- Once the objective is clear , other activities in planning remain focused around it.

PLANNING A WEB SITE

- **According to James Lewis following must be objectives:**

1. Web projects must be specific.
2. The Web project should be Measurable.
3. The schedule of the web projects must be Attainable.
4. There should be Realistic approach for deciding the web project.
5. It should be Time-limited or should not continued for a very long period.

PLANNING A WEB SITE

- **Goal** means the achievement from the development of the website.
- It depends on the reason for the development.
- If a site is developed to provide the information to the users, the goal should be user satisfaction in terms of accurate information.
- If it is business website, goal is to increase in the sell.
- For product support Web sites, better services to the customer is the goal.
- Both **objective** and **goal** of the Web site development should be clear, well-documented form.

PLANNING A WEB SITE

❑ Audience/ User Profile

- Very important to know about background, education, gender, age, liking and preferences of target audience.
- This will lead us to choose not only the look and feel of the web site, but also make it simple to choose the contents and their organization as well as navigation styles.
- Asking the questions such as
  - Who are the visitors?
  - Why do they visit the site?
  - How frequently do they visit?
  - What are their expectation, liking and preferences

PLANNING A WEB SITE

- **Ways to find out the target audience**

- **Market research**
  - Help you to understand exactly what kind of audience is accessing your web page and is its demand.
  - Many firms which provide these kind of services to web companies.
  - Hence your web company can give a contract to such firms for making the market research.

PLANNING A WEB SITE

- **Focus groups**
  - Getting service for market research from some private firms is costly.
  - Hence we can use another approach of identifying users.
  - The focus groups is a group of people who represent the target users.
  - The web companies can make use of focus group for getting the feedback on proposed ideas on the website.
  - This group can also give the comment on how target audience think on the ideas that are put on the website.

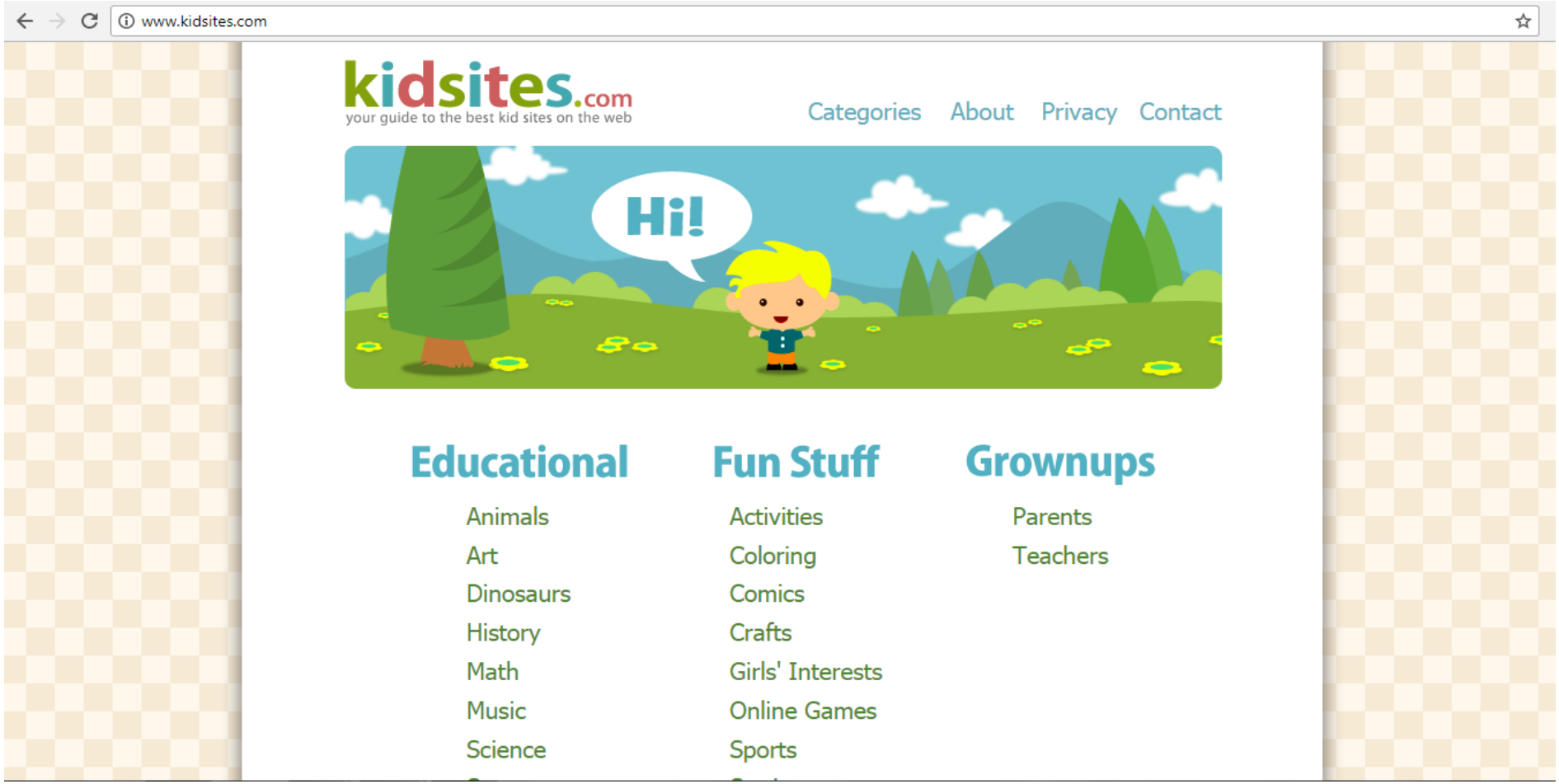
PLANNING A WEB SITE

- **Understand intranet audience**
  - If your website is created for an intranet access is for several departments.
  - Every department has several needs.
  - Hence becomes very easy and manageable to understand the target audience for intranet.

PLANNING A WEB SITE

❑ Some Examples

- For children website, we can use cartoons and graphics with colors.



PLANNING A WEB SITE

Some Examples

- For users who are technology savvy, we can use advanced technology and features, as they likely using latest software.
- For users interested in music we can use background music.
- For the common users , we have to restrict to simple and easy organization of contents so that they can find desired contents with little effort.

PLANNING A WEB SITE

Identifying and Organizing Contents

- Basic objective for any web site is to provide the information in one or other form such as text, images, audio, video, various forms etc.
- This information forms the contents of the web site.
- The content of the web site is collected from the various sources depending upon the objective of the web site and user needs.
- For example, the contents for the web site of tax department are the rules and regulations and the various forms to download which the citizens can fill and submit.

PLANNING A WEB SITE

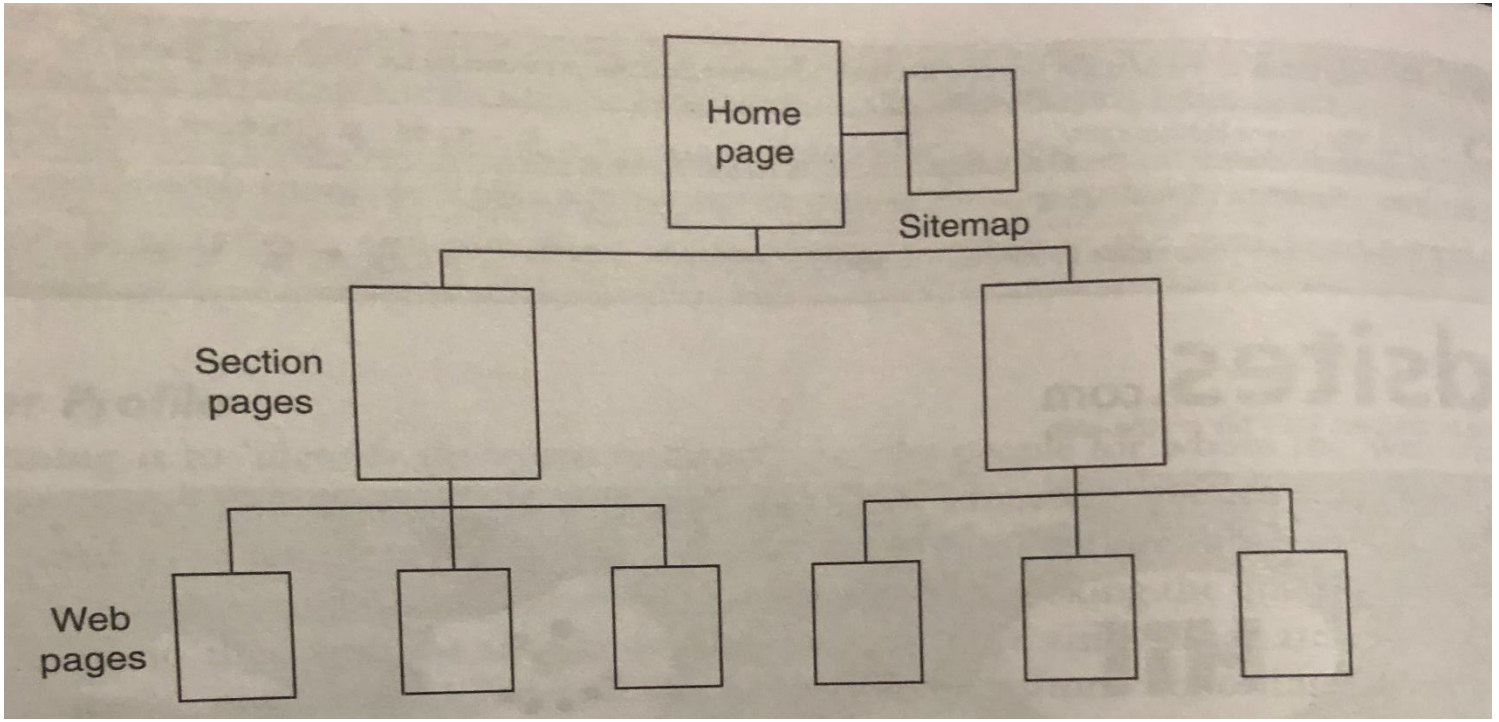
- **Important characteristics**

- Contents includes text, graphics, forms, sound and video.
- Contents should be unambiguous, clear, spell checked, precise and accurate.
- Content should be relevant, recent and matching to site objective.
- Content should fulfill the information need of the users.
- Content should be well organized.

PLANNING A WEB SITE

- Once the information is identified and collected, the next important task is to prepare a story board that includes the page layouts, their logical grouping and navigation choices.

- Either paper-based approach is used or we can use the flow chart software to prepare it.



PLANNING A WEB SITE

- For the Web site with large information base, search option is provided on the top of each page to search the desired information quickly.

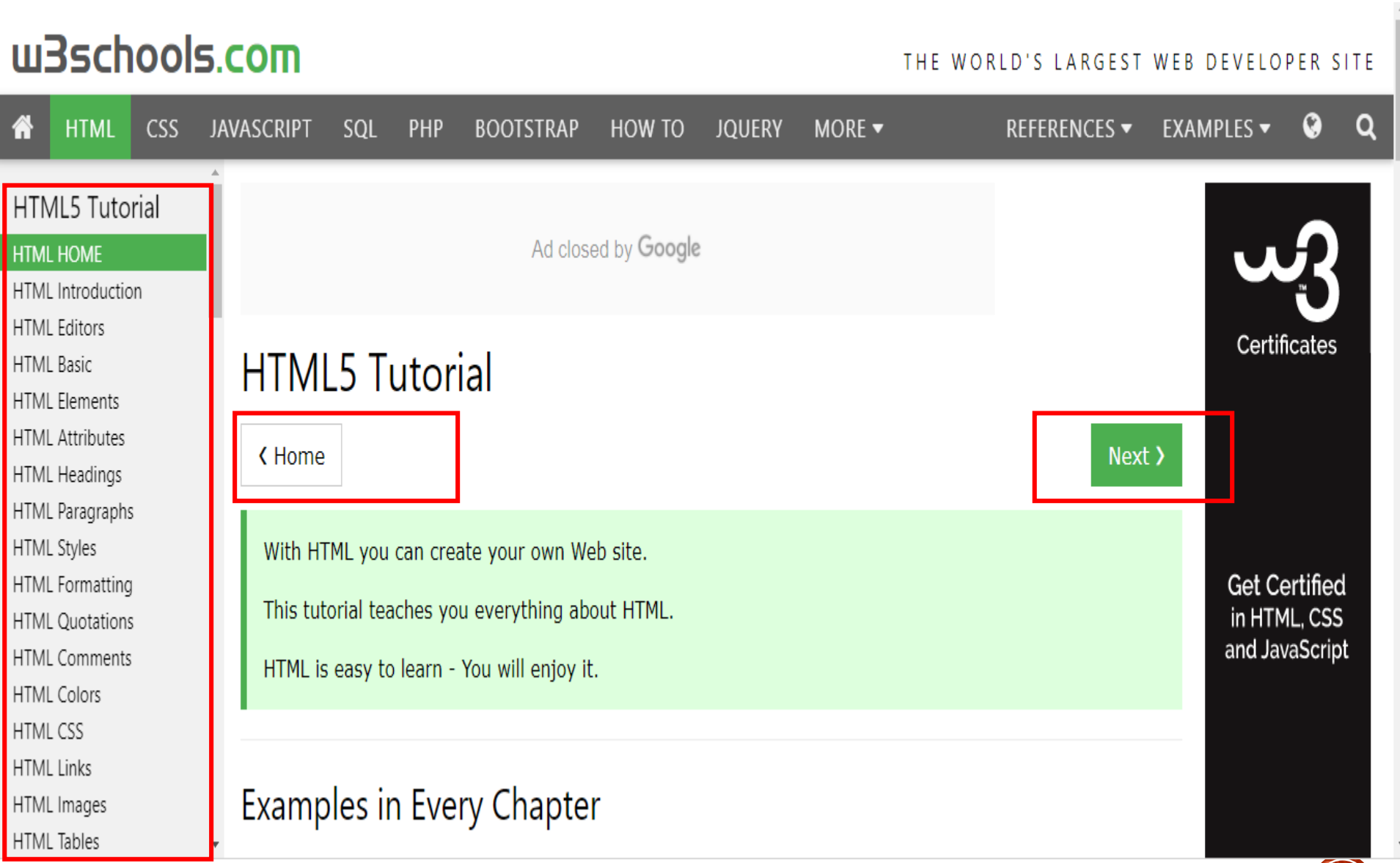
- Navigation menu on left-hand side or on the top helps user move to desired section directly.

PLANNING A WEB SITE

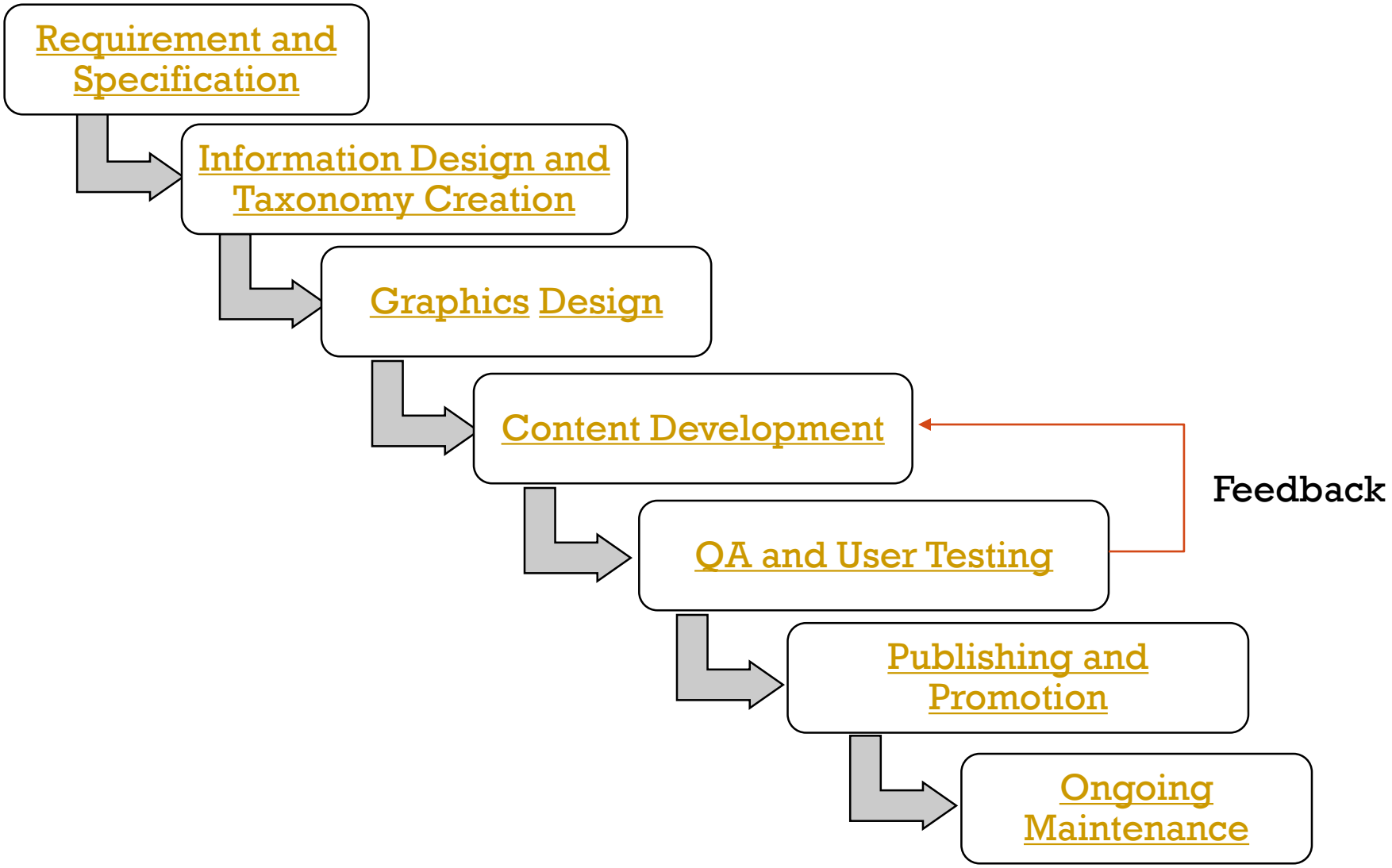
- **Example**

- For online tutorial Web site, home page list all the topics.
- Once the topic is chosen, a topic list as navigation menu is provided on first page and then sequentially or randomly each page can be visited.
- Each page of tutorials provides the ‘previous’ and ‘next’ link to move in both direction while reading tutorials.
- Search facility cab be provided on each page.

PLANNING A WEB SITE



WEB SITE DEVELOPMENT PROCESS



WEB SITE DEVELOPMENT PROCESS

Requirements and Specification

- Requirements are collected from the clients.
- Nothing but the list of customer needs.
- It may include search facility, specific navigations, particular color, branding requirement and so on
- These requirements are then analyzed and web development team prepares project specification that contain design requirements, page layout sketches, technical requirements and so on.

WEB SITE DEVELOPMENT PROCESS

Information Design and Taxonomy Creation

- The designers prepare the structure of the site in such away that the contents can be represented in most meaningful way and easy to navigate by target user.

- During this stage, the taxonomy of the site information is developed.

- Taxonomy is for giving the section names and names to navigation links and menu system of the site.

WEB SITE DEVELOPMENT PROCESS

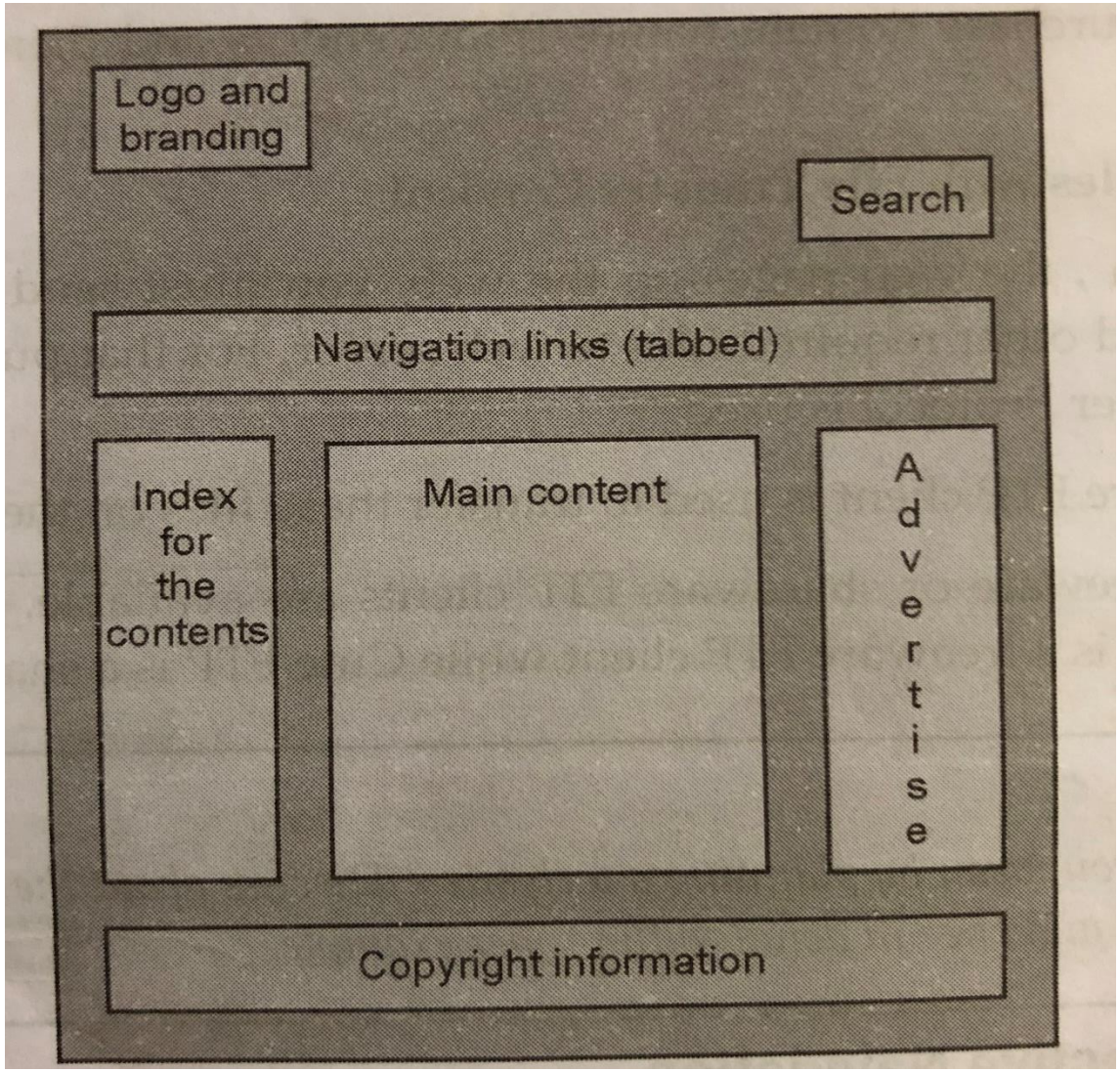
Graphics Design

- At the same time of information design and taxonomy creation the design sketches and page mockups can be prepared for the site.

- These mockups can be easily modified based on the suggestions to create the web page design.

WEB SITE DEVELOPMENT PROCESS

- **Web page Mockup**



WEB SITE DEVELOPMENT PROCESS

❑ Content Development

- All the technical activities such as
  - content development
  - coding and
  - validation are conducted.

WEB SITE DEVELOPMENT PROCESS

Quality Assurance and User Testing

- The development team performs various test such as
  - cross browser compatibility,
  - connectivity at different bandwidths
  - testing links
  - data forms and
  - multimedia technologies.

WEB SITE DEVELOPMENT PROCESS

- **Publishing and Promotion**

- Website is published to the web or organizations intranet.
- The client then begins to make the promotion of this site.

- **Ongoing Maintenance**

- The web contents must be updated and kept upto date
- New sections can be added based on the requirement.
- The newer interactive features can be added to make the site more interactive.

PROCESS OF PUBLISHING WEB SITE

Step 1: Choose Web Hosting Service Provider

- The service provider is basically a company that hosts web pages on web server, making them available to anyone who knows the URL of the web site.
- The web hosting service can be available for hosting both personal as well as business web sites.
- The web host provide the internet access, Email accounts and space for personal or business web site.

PROCESS OF PUBLISHING WEB SITE

Step 2: Register domain name

- The domain name basically indicates the actual location of the web site files on the web server.
- The domain names are managed by Internet Corporation for Assigned Names and Numbers (ICANN).
- You can purchase domain name from vendors and then they register it with ICANN.

PROCESS OF PUBLISHING WEB SITE

Step 3: Upload files with File Transfer Protocol

- To publish, the web pages on the web, you must send web documents, images and other required files to web server.
- For that purpose FTP(File Transfer Protocol) is used.
- FTP client is used to transfer these files on the web page.
- Various freeware or shareware FTP clients are available.
- For example Filezilla or FireFTP is a freeware FTP client while Cute FTP is shareware FTP client.

NAVIGATION

- The ways to move from one page to another page in a website using hyperlinks provided on a page.
- If navigation design is not proper then user feels the problem in moving around the pages in the site in a desired manner or gets confused and leaves the site.
- The hyperlinks allows the user to navigate from one page to another.
- Certain color of the hyperlinks indicates whether the corresponding web page is visited or not. The unvisited links are in one color and after using the hyperlinks their color gets changed.
- This feature helps the user to understand where they are and what have they visited.

NAVIGATION

- **Some important tips to design effective navigation**

- Navigation links are either text based, i.e. word or phrase is used as a link, or graphical, i.e. a image , a icon or a logo is used as a link.

- Navigation link should be clear and meaningful. Use the word related to section or information. If graphical link is used, picture should be meaningful.

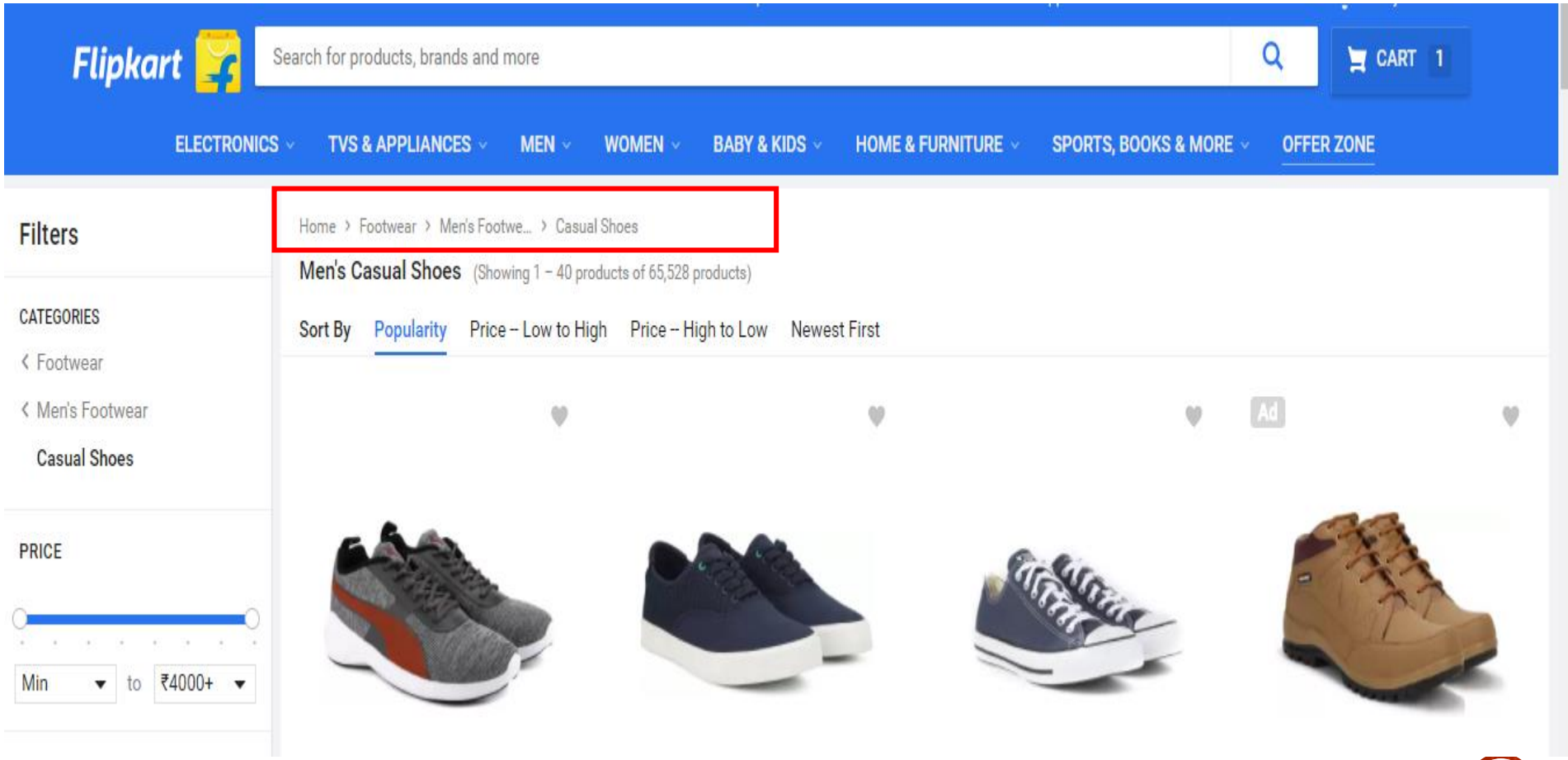
- It should be consistent, i.e. use same types and color on each page. For example, ‘sign out’ link can be placed on each page on top right-hand side with same color and type.

NAVIGATION

- Link should be understandable. Link itself guides the user about the contents.
- Provide search link, if necessary, usually on top of the page. Use some common links such as ‘About us’ or ‘Contact us’. Such links should appear on each page at same place, normally on navigation menu or horizontal navigation bar.
- Provide the way to return to first page from anyware using link ‘Home’ so incase of confusion user can start from home page.

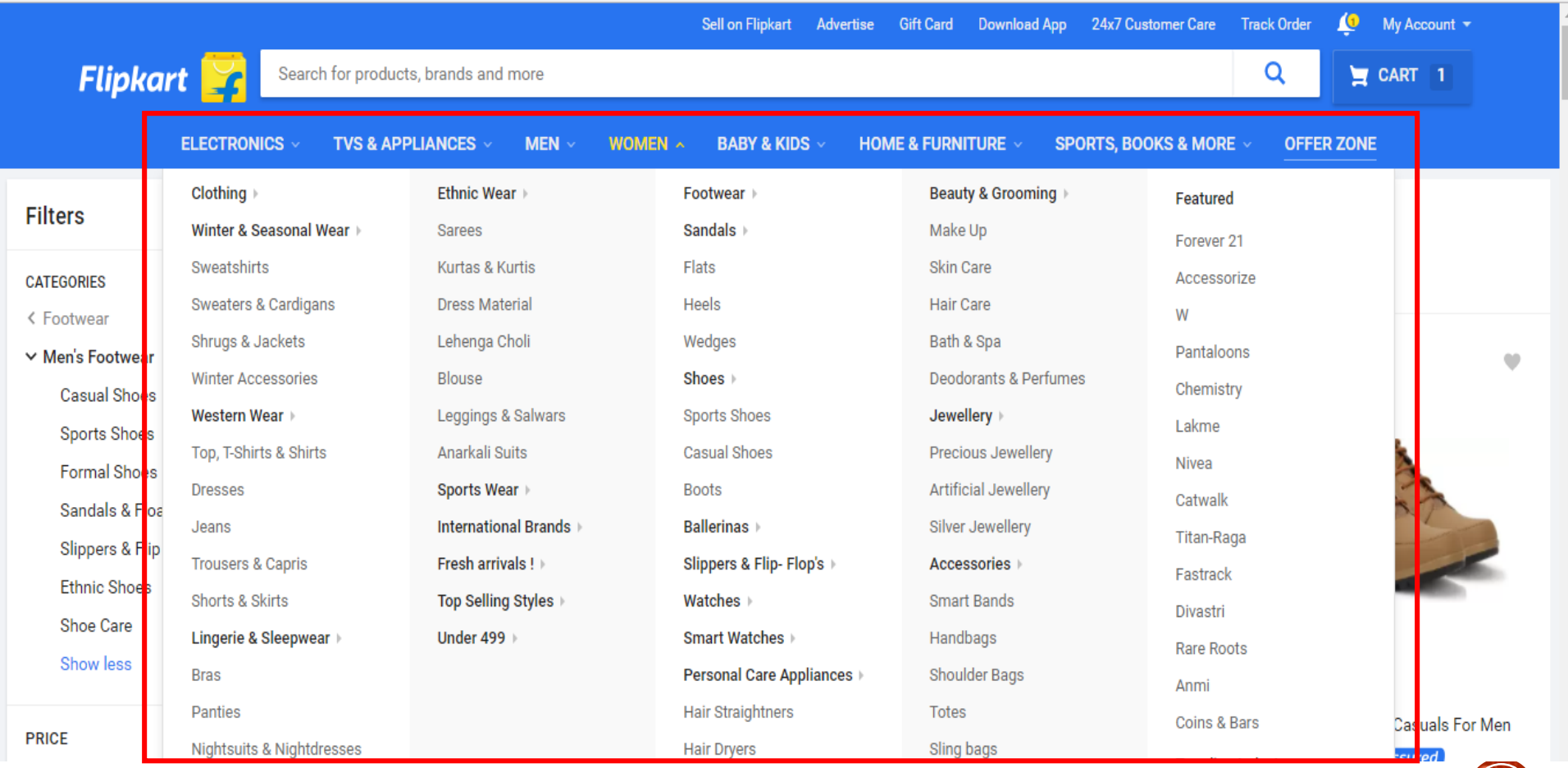
NAVIGATION

- Provide the user with information regarding location, i.e. where is she/he?



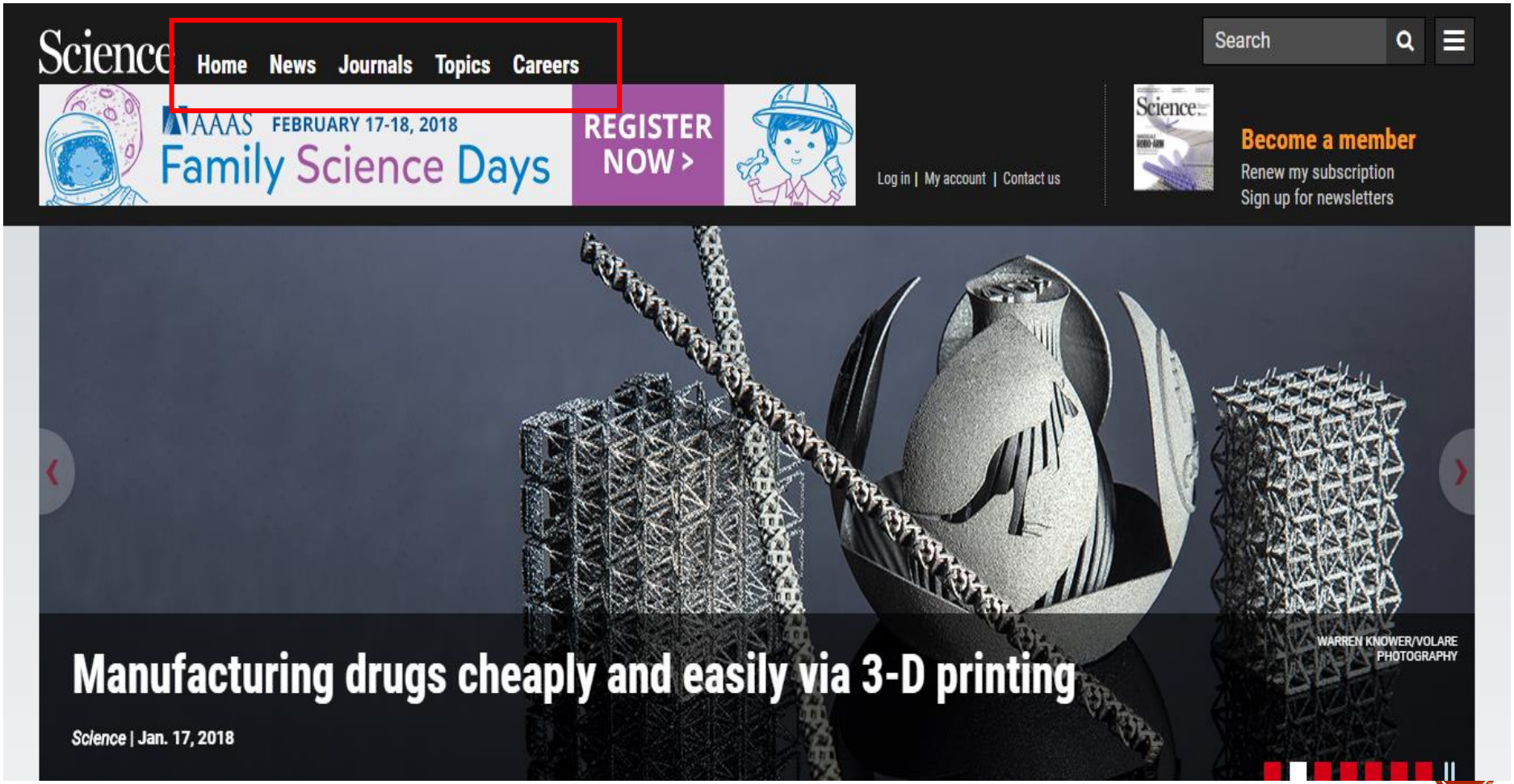
NAVIGATION

- For website having many clear sections, provide navigation menu to directly move to a particular section.



NAVIGATION

- Horizontal navigation bar can be provided on each page to directly jump to any section from anyware.



END